

Fractional Follies—Solutions

Fall 2020 ARML Power Contest

Problem 1. Apply the rules for generating a sequence from a program and a starting value:

- (a) Since $\frac{4}{3} \cdot 2$ is not an integer, but $\frac{3}{2} \cdot 2 = 3$, the next term $n_1 = 3$. Then $\frac{4}{3} \cdot 3 = 4$, then $\frac{3}{2} \cdot 4 = 6$. Continuing, the first eight terms generated are: $\boxed{3, 4, 6, 8, 12, 16, 24, 32}$.
- (b) Similarly we obtain $\boxed{5, 36, 858, 234, 5577, 1521, 3549, 8281}$.

Problem 2. If $b > 0$ then $\frac{2}{3} \cdot 2^a 3^b = 2^{a+1} 3^{b-1}$, while if $b = 0$ then $\frac{2}{3} \cdot 2^a$ is not an integer and the program terminates. So as long as $b > 0$ the program continues to decrease b and increase a by the same amount until the exponent on the 3 reaches 0, at which time the exponent on the 2 will reach $a + b$. That is, the program terminates at $\boxed{2^{a+b}}$.

Problem 3. This time, the program deducts one from each of the exponents on the 3 and the 5, adding one to the exponent of the 2, until either a or b is reduced to zero.

First, if $a \geq b$ then reducing both a and b by one b times will reduce b to zero, while reducing a to $a - b$. In this case, the final result will be $2^b 3^{a-b}$. Similarly, if $a \leq b$ then the final result will be $2^a 5^{b-a}$. Of course, if $a = b$ these results agree. Putting these together,

the final result is $\boxed{\begin{cases} 2^b 3^{a-b} & a \geq b \\ 2^a 5^{b-a} & a \leq b \end{cases}}$.

Problem 4. Building on the result of the previous problem, note that if $a \geq b$ then $c = \max(a, b) = a = b + (a - b)$. Since the result of the program $\left(\frac{2}{15}\right)$ was to leave $2^a 3^{b-a}$ all that remains is to add the exponent on the 3 to the exponent on the 2; this was accomplished by the program $\left(\frac{2}{3}\right)$. So do this program after doing the $\left(\frac{2}{15}\right)$ program to achieve the desired result.

On the other hand, if $a \leq b$ then $\max(a, b) = b = a + (b - a)$. In this case, we have to add the exponent remaining on the 5 to the exponent on the 2. This would be accomplished by the program $\left(\frac{2}{5}\right)$.

Since $\left(\frac{2}{3}\right)$ has no effect on numbers where the exponent on the 3 is zero (and similarly for $\left(\frac{2}{5}\right)$ for the exponent on the 5), these can be combined into one program: first move equal amounts from the exponents on the 3 and the 5 to the 2 until one of the exponents is reduced to zero, then move any remaining amount on the other exponent to the 2. Thus, two programs that accomplish the desired task are $\boxed{\left(\frac{2}{15}, \frac{2}{3}, \frac{2}{5}\right)}$ and $\boxed{\left(\frac{2}{15}, \frac{2}{5}, \frac{2}{3}\right)}$.

Problem 5. Since $k \neq 1$ then $b \neq a$ (unless $a = 0$; this exception was not stated in the problem—the word “positive” should have been required of the input, n_0 , to a program!). So at least one of the fractions of the program must multiply a to give an integer, otherwise the program would terminate at $a \neq b$. But if $f_i a$ is an integer, a' , then $f_i b = f_i k a = k a'$ is an integer. One of the fractions in the program before f_i might be applied to b , but if none do than certainly f_i does. In either case, the program does not terminate at b .

Problem 6. The simplest approach is to use the example in the paragraph preceding problem 5 as a model. For example, one such program is $\left(\frac{11}{6}, \frac{2}{11}, \frac{13}{8}, \frac{4}{13}, \frac{17}{9}, \frac{3}{17}\right)$.

Problem 7. If $n = 2^a 3^b$ and both a and b are positive, then P applied to n yields $2^{a-1} 3^{b-1}$. If either a or b is zero, the program terminates. Thus, if $a \leq b$ then the program acts on n_0 a times, reducing both a and b by a , leaving $2^0 3^{b-a}$. On the other hand, if $a > b$ then the program acts on n_0 b times before the exponent on the 3 is reduced to zero, leaving $2^{a-b} 3^0$. Since $b - a = 0$ when $a = b$ the final numbers shown are as described in the problem.

Problem 8. The key to this problem is to reduce the exponent on the 3 by two, maintaining its parity. When it can no longer be reduced, the remaining number will be either 3^0 if b was initially even, or 3^1 if b was originally odd. At that point, the exponent on the 3 should simply be transferred to the 2. These tasks are accomplished by the program $(\frac{1}{9}, \frac{2}{3})$. As long as the exponent on the 3 is two or larger, the $1/9$ reduces it by two. Then, as shown above, the $2/3$ adds the exponent on the 3 to that which is on the 2, completing the desired program.

Problem 9. First, if $b = 0$ so that $n_0 = 3^0 = 1$ then none of the fractions in P will multiply n_0 to give an integer, so the program terminates with result 1 as desired.

Next, if $b \geq 1$ the first iteration of the program produces $3^{b-1} 5$. Now as long as $k > 0$ the number $3^k 5$, when acted upon by the program, produces $3^{k-1} 7$ then $3^{k-1} 5$. In other words, after two steps the number produced reduces the exponent on the 3 by one, while leaving the 5 alone. Thus, after $2k$ steps (by induction, if you like) the exponent on the 3 is reduced to zero, leaving just 5. Of course, the next step then produces 2, and terminates there. Setting $k = b - 1$ thus shows that starting with 3^b with $b > 0$, after $1 + 2(b - 1) + 1$ steps the program terminates at 2.

Problem 10. One idea is to subtract the smaller of b or c from both exponents. This leaves $3^0 5^0$ if $b = c$, 3^{b-c} if $b > c$ and 5^{c-b} if $b < c$. Then the technique of the previous problem tests which, if either, exponent remains positive, putting the correct number into the exponent of the 2. One such program is $P = (\frac{1}{15}, \frac{11}{21}, \frac{7}{11}, \frac{7}{3}, \frac{2}{7}, \frac{17}{65}, \frac{13}{17}, \frac{13}{5}, \frac{4}{13})$.

This program works by first applying $\frac{1}{15}$ until the number becomes $3^{b-c} 5^0$ if $b \geq c$ and $3^0 5^{c-b}$ if $b < c$. If $b = c$ the program terminates here, at 1. If $b > c$ then the next four fractions emulate the previous problem (with 5 and 7 replaced by 7 and 11, respectively, since 5 is already being used in this problem), to terminate at 2. In case $b < c$ the final four fractions apply instead, emulating the previous problem (with 13 and 17 taking the places

of 5 and 7), and the program terminates at 4 due to the numerator of the very last fraction in the program.

Problem 11.

- (a) If $a = 0$ then $n_0 = 3^b$. Since every fraction in the program has a prime in its denominator other than 3, only the last fraction can be applied, and this never changes. Thus each iteration through the program reduces the exponent on the 3 by one until the terminal number $3^0 = 1$ is reached. This is the desired result.
- (b) If $b > 0$ then the next number produced is $2^{a-1}3^{b-1} \cdot 5 \cdot 7 \cdot 13$. This is then converted to $2^{a-1}3^b - 1 \cdot 5 \cdot 7 \cdot 11$. If the exponent on the 3 continues to remain positive, the first two fractions are applied again, to arrive at $2^{a-1}3^{b-2} \cdot 5^2 \cdot 7^2 \cdot 11$. The first two fractions continue to reduce the exponent on the 3 by one while increasing the exponents on the 5 and the 7 by one. So after $2b$ steps the sequence now has the value $2^{a-1}5^b7^b \cdot 11$. Finally, the $\frac{1}{11}$ is applied to make the result at this stage $2^{a-1}5^b7^b$.
- (c) Since there is now no factor of 11 in the current number, but there are factors of 7, the $\frac{3}{7}$ will be continually applied until the exponent on the 7 is reduced to zero and the program can once again access $\frac{11}{2}$ to obtain a factor of 11. Of course, applying $\frac{3}{7}$ repeatedly simply moves the exponent currently on the 7 to the 3. So after this phase is complete, the sequence stands at $2^{a-1}3^b5^b$.
- (d) Given that the program is initiated with the number $n_0 = 2^a3^b5^0$, the previous three parts can be assembled into an induction proof that the terminal number is 5^{ab} .

Base: if $a = 0$ part (a) shows that the final result is 5^0 as required.

Induction: Assume that, if initiated with $2^a3^b5^k$ for some arbitrary $k > 0$, that the program terminates with 5^{ab+k} . Consider the program initiated with $2^{a+1}3^b5^k$. By part (b), before any fraction other than the first three in the program are employed, the sequence has a current value of $2^a5^{b+k}7^b$. Part (c) shows that this is then converted to $2^a3^b5^{b+k}$. By induction, the program now terminates with a value of $5^{ab+(b+k)} = 5^{(a+1)b+k}$. Setting $k = 0$ completes the proof.

Problem 12. First, if $a = 0$, the Hamming weight of a is $h = 0$ while for $a > 0$ the Hamming weight $h(a)$ satisfies the recursion

$$h(a) = \begin{cases} h(\lfloor a/2 \rfloor) + 1 & a \text{ is odd} \\ h(\lfloor a/2 \rfloor) & a \text{ is even} \end{cases}.$$

This is true because for any a , the Hamming weight of a and of $2a$ are equal, since multiplying by two merely adds a zero to the (left) end of the binary expansion of a , while the number $2a + 1$, when expressed in binary, ends in a 1, and thus has a Hamming weight one more than that of $2a$. In short, $h(a) = h(\lfloor a/2 \rfloor) + p$ where p , the parity bit, is 0 if a is even and 1 if a is odd.

To prove the program behaves as required, the proof will go slightly further. It will show that if the program is started with the input 2^a13^k it terminates with $2^0 \cdot 13^{h(a)+k}$ where $h(a)$ is the Hamming weight of a . The problem will then follow from setting $k = 0$.

First, if $a = 0$ then $n_0 = 2^0 \cdot 13^k = 13^k$. Since every fraction in the program has a prime other than 13 in its denominator, no term in the program produces an integer on multiplication by 13^k so the output is also 13^k and the program terminates. So the program behaves as promised for $a = 0$.

If $a = 1$, the Hamming weight $h(1) = 1$, and $n_0 = 2 \cdot 13^k$. The sequence produced by the program starting from this n_0 is $2 \cdot 13^k, 7 \cdot 13^k, 10 \cdot 13^k, 13^{k+1} = 13^{h(a)+k}$ where it terminates, as required. So the program exhibits the desired behavior for $a = 0$ or $a = 1$.

Now proceed by induction on the number, s , of binary digits in a . The work above shows base case $s = 1$ to be true. So assume that the program behaves as desired for all numbers with s binary digits, and let a have $s + 1$ binary digits. The first few steps of the program will produce $2^a \cdot 13^k, 2^{a-1} \cdot 7 \cdot 13^k, 2^a \cdot 5 \cdot 13^k$.

At this point, the first two fractions in the program alternately operate on the number. The first fraction produces $2^{a-2} \cdot 3 \cdot 11 \cdot 13^k$ then the second produces $2^{a-2} \cdot 3 \cdot 5 \cdot 13^k$. These first two fractions continue to operate, reducing the exponent on the 2 by two and increasing the exponent on the 3 by one. This happens $\lfloor a/2 \rfloor$ times until the exponent on the 2 is either zero or one. The number at this point is now

$$2^{a-2\lfloor a/2 \rfloor} \cdot 3^{\lfloor a/2 \rfloor} \cdot 5 \cdot 13^k.$$

- if a was initially odd, the current number is $2 \cdot 3^{\lfloor a/2 \rfloor} \cdot 5 \cdot 13^k$. The next number in the sequence is $3^{\lfloor a/2 \rfloor} \cdot 13^{k+1}$. The fraction $2/3$ now repeatedly acts on the number to turn it to $2^{\lfloor a/2 \rfloor} \cdot 13^{k+1}$ before any other fraction will be able to act on the number.
- if a was initially even, the current number is $3^{\lfloor a/2 \rfloor} \cdot 5 \cdot 13^k$. The next fraction to act is the $1/5$, eliminating the factor of 5. Then, as in the previous bullet point, the $2/3$ acts to yield $2^{\lfloor a/2 \rfloor} \cdot 13^k$.

So the number $2^a \cdot 13^k$ has been changed into $2^{\lfloor a/2 \rfloor} \cdot 13^{k+p}$ where p is 0 if a was even and 1 if a was odd. Since $\lfloor a/2 \rfloor$ has only s binary digits the inductive hypothesis implies that the final output of the program will now be $13^{k+h(\lfloor a/2 \rfloor)+p}$. By the recursive formula for $h(a)$ given above, this is precisely $13^{k+h(a)}$ and the proof is complete.

The fractional programs described in this Power Problem are examples of the **FRACTRAN** programming language invented by John H. Conway around 1987. Conway was one of the most inventive mathematical minds of the past several decades, dabbling in many branches of mathematics. He invented the game of Life, an efficient way of describing knotted strings, and surreal numbers, among many other discoveries. He died in April of this year of complications from COVID-19.

The **FRACTRAN** programming language is actually *Turing complete*. That means that any computation that can be done on any (non-quantum) computer in any computer language can be performed by a **FRACTRAN** program. More details can easily be found on the Web or in Conway and Guy's *Book of Numbers*, ISBN 0-387-97993-X.